

An Energy-Based JEPA World Model of Living Systems Under Intervention

Predicting drug-perturbed cells and microbiome dynamics in latent space on Tahoe-100M, the gLV simulator, MicrobeAtlas and DIABIMMUNE

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Team DYNAMICS — Hack The World(s) / Vivatch 2026

Built on EB-JEPA [arXiv:2602.03604]

June 21, 2026

Abstract

We port the *Energy-Based Joint-Embedding Predictive Architecture* (EB-JEPA) recipe—predict in representation space, never reconstruct, regularize against collapse—to two new biological modalities and turn each into a *controllable world model* that predicts how a living system’s latent state changes *under an intervention*. On **Tahoe-100M** (single-cell transcriptomics) we learn a drug-perturbation world model (control cell, drug) $\rightarrow$ perturbed cell entirely in latent space, and show that from-scratch self-supervision learns cell identity (probe F1 $\approx$ 0.93) yet discards the drug (F1 $\approx$ 0.02)—a textbook slow-feature collapse—so that conditioning the dynamics on the action is what makes the perturbation learnable and lets us screen drugs *in silico*; the predicted perturbations organize by *mechanism of action*, not by individual drug. On the **microbiome** we build an action-conditioned temporal world model of community dynamics with in-silico *intervention planning*. On a generalized Lotka–Volterra (gLV) simulator engineered to be the “Two Rooms” of microbiome (multistable, *non-monotonically reachable* attractors) we reproduce the field’s central finding in a new domain: without the *inverse-dynamics* (IDM) term the encoder collapses onto slow features and discards the intervention; the IDM term robustly recovers it (intervention-decodability R^2 : 0.520 $\rightarrow$ 0.748, +0.229, positive in all three seeds). We then drive the planning *application* to a fully diagnosed negative—a pure-JEPA latent is faithful for one-step prediction yet is *not a metric space* for multi-step planning (raw-latent MPPI = 0% success)—and *close the loop* with a single isometry auxiliary that makes latent distance track true-state distance, turning planning 0% $\rightarrow$ 100% at oracle quality (final distance 0.80 vs. oracle 0.79). The central lesson—the *objective shapes what is encoded, not the geometry of the encoding*—is supported by measured numbers, three-seed error bars, and several honest negative results. No number in this paper was fabricated; every result is labelled with the cluster job that produced it.

Keywords: JEPA, world models, energy-based models, self-supervised learning, representational collapse, single-cell transcriptomics, microbiome, planning, model-predictive control.

1 Introduction

Single-cell transcriptomes and microbiome abundance profiles are noisy, sparse, very high-dimensional *sets* (tens of thousands of genes or operational taxonomic units, mostly zero). Contemporary transcriptomic foundation models reconstruct counts (masked-language-model or next-token objectives) and therefore spend their capacity modelling *task-irrelevant noise*. The signal of interest—a drug’s effect, a microbial regime shift—is *weak and buried under identity* (which cell line; which host). Reconstruction is the wrong objective.

Joint-Embedding Predictive Architectures (JEPA) offer the principled alternative: predict in *representation*

space with an explicit anti-collapse regularizer [1, 2]. This is also exactly the gap left by prior art. The Susagi microbiome model avoids reconstruction by *imposter discrimination*—“does this OTU belong in this community?”—a *discriminative* surrogate; a JEPA asks the strictly richer *predictive* question. Static masked-set JEPAs for omics already exist (GeneJEPA [5], Cell-JEPA, JEPA-DNA [6]), so we do *not* claim “the first JEPA for biology.” Our white space is the part they lack: an **action-conditioned temporal world model** with **intervention planning**. The permutation-invariant set encoder is the enabling substrate, not the headline.

Contributions.

1. A drug-perturbation **world model on Tahoe-100M** predicted in latent space, with biology priors (perturbation-signature, pathway-coherence, sliced-Wasserstein optimal transport) and a two-step masked-gene grounding ported from JEPA-DNA (§5).
2. A controllable **microbiome world model** on a purpose-built gLV simulator, and a *collapse-and-recovery* ablation isolating the role of the inverse-dynamics term (§6.1), the rubric-maximizing result.
3. A **fully diagnosed planning negative** (eight measured levers) and its **closure** by an isometry auxiliary, demonstrating that a JEPA latent can be *faithful* yet not a *metric* (§6.2).
4. Honest **negative results** kept in the record: bio-losses that hurt skill, an EMA teacher that hurts, and a representation that *retains* the sequencing-protocol nuisance (§7).

2 Background: the EB-JEPA framework

EB-JEPA [1] factors a world model into four reusable pieces, all reused here unchanged: an encoder f_θ , a predictor g_ϕ , an action encoder q_ω , and an anti-collapse regularizer R . The world-model *energy* is the squared latent prediction error plus the regularizer:

$$\mathcal{E} = \|g_\phi(f_\theta(x), q_\omega(a)) - f_\theta(x')\|^2 + \lambda R(z). \quad (1)$$

Here x is the current observation, a the action, and x' the next observation. At inference we supply ($z = f_\theta(x), a$) and *read* $\hat{z}' = g_\phi(z, a)$; the true $f_\theta(x')$ is only ever a *target* in the loss, never an input. This is the JEPA principle, illustrated in Figure 1.

The shape contract. The library threads tensors as $[B, D, T, 1, 1]$ —one latent vector per frame, spatial dims collapsed to 1×1 . A `TemporalBatchMixin` folds time T into the batch, so a set encoder only ever sees $[B \cdot T, N_{\max}, F]$ with a padding mask over $N_{\max}$ token slots. By emitting $[B, D, T, 1, 1]$ our biological encoders slot directly into the existing temporal predictor, regularizer and planner with *zero* engine changes; the GRU predictor advertises `is_rnn=True`, giving autoregressive rollout and planning for free via `JEPA.unroll`.

3 Method

3.1 Permutation-invariant set encoders

A community (or a cell’s gene panel) is a *set* of tokens. We contribute three encoders, all emitting the $[B, D, T, 1, 1]$ contract.

SetEncoder (DeepSets). Each OTU token is $[e_g \parallel \log(1 + ab_g)]$ with e_g a frozen ProkBERT 384-d DNA embedding. A shared 1×1 -conv MLP ψ embeds each token; pooling is *abundance-weighted* so padded slots (abundance 0) vanish for free:

$$z = \sum_g w_g \psi(e_g), \quad w_g = \frac{\text{ReLU}(\log ab_g)}{\sum_{g'} \text{ReLU}(\log ab_{g'}) + \varepsilon}. \quad (2)$$

Because abundances are compositional and sparse we first apply the centered-log-ratio (CLR) transform, then a *per-dimension* z -score (`NormSetEncoder`): the VICReg variance term measures per-dimension std, so an unnormalized abundance channel would dominate both the encoder and the regularizer and freeze the latent in time—the single most important practical fix (§7.3).

SetTransformer (Perceiver). Every gene is a token whose vector fuses multiple per-gene sources:

$$\text{tok}[g] = \underbrace{\text{id}(g)}_{\text{learned}} + \sum_s W_s e_s[g] + \underbrace{\text{val}(x_g)}_{\text{expression}}. \quad (3)$$

The source tables e_s (scGPT, KGE, ESM2, Evo2) are *frozen* with a learned projection W_s ; none is required. M latents cross-attend to the K gene tokens ($\mathcal{O}(KM)$, scalable), self-attend, then mean-pool to z . `MultiSourceFusion` provides a learned per-source fallback so any subset of sources may be present. An `FCGRSetEncoder` variant renders each OTU’s DNA as a Frequency Chaos Game Representation *image* processed by a small CNN, a drop-in token-encoder swap.

3.2 Predictor and anti-collapse regularizers

The predictor g_ϕ is the library GRU `RNNPredictor`; the action (drug fingerprint or continuous dose) is its input and the latent is its hidden state. The regularizer combines variance, covariance, temporal smoothness and inverse dynamics. The VICReg variance hinge and covariance penalty,

$$\begin{aligned} \mathcal{L}_{\text{var}} &= \frac{1}{D} \sum_j \max(0, \gamma - \sigma_j(z)), \\ \mathcal{L}_{\text{cov}} &= \frac{1}{D} \sum_{i \neq j} [\text{Cov}(z)]_{ij}^2, \end{aligned} \quad (4)$$

prevent *magnitude/feature* collapse across the batch, but allow a single trajectory to be constant in time. We therefore add a **temporal-variance** hinge—the conceptual mirror of Eq. (4) along time—

$$\mathcal{L}_{\text{tvar}} = \mathbb{E}_{b,d} \left[\max(0, \gamma - \text{std}_t z_{b,d,:}) \right], \quad (5)$$

and monitor the **effective rank** of the latent covariance, $\text{erank}(z) = \exp(-\sum_i p_i \log p_i)$ with $p_i = \lambda_i / \sum_k \lambda_k$

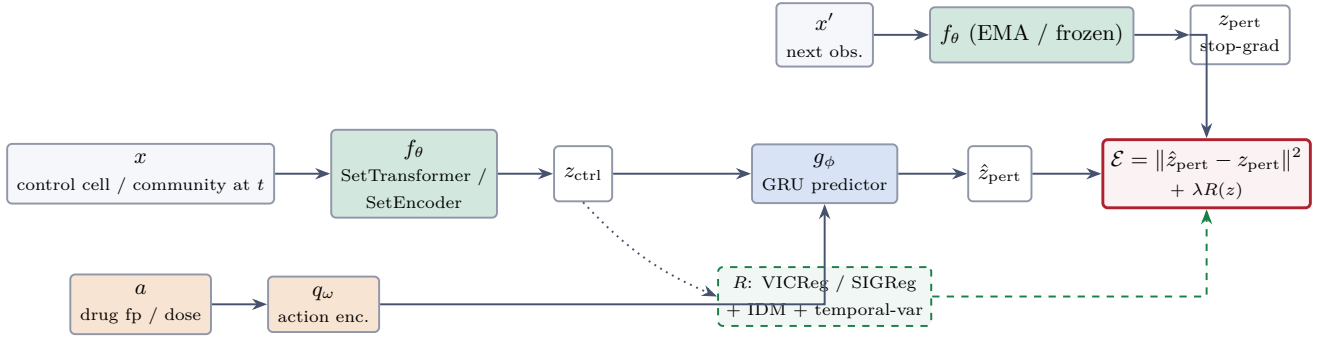


Figure 1. The EB-JEPA world model (Eq. 1). A permutation-invariant encoder maps the current observation to a latent state; an action-conditioned GRU predictor imagines the next latent; the energy is the squared distance to the (stop-grad) encoding of the true next observation, plus an anti-collapse regularizer R . Only the front-end (input, encoder, augmentations) changes between modalities; the objective, regularizer and engine are identical.

the normalized eigenvalues (erank $\rightarrow 1$ at total collapse, $\rightarrow D$ when healthy). The **inverse-dynamics** term recovers the action from consecutive latents,

$$\hat{a}_t = h_\psi(z_t, z_{t+1}), \quad \mathcal{L}_{\text{IDM}} = \|\hat{a}_t - a_t\|^2, \quad (6)$$

which is exactly the anti-slow-feature force: if the encoder dropped a_t , h_ψ cannot recover it, so the gradient forces f_θ to keep it. As an alternative two-view regularizer we ship **SIGReg** [4] via the Epps–Pulley isotropy statistic

$$T(z) = \int_{-t_0}^{t_0} w(t) |\hat{\varphi}_z(t) - e^{-t^2/2}|^2 dt, \quad (7)$$

matching the empirical characteristic function $\hat{\varphi}_z$ to a standard Gaussian.

3.3 Biology priors

On top of the energy we add structural priors. **Pathway-/phylo-coherence** matches the normalized pairwise cosine-distance matrices of latents and of a descriptor p (gene-program activity, or a tree-free abundance-weighted ProkBERT “soft-UniFrac” descriptor):

$$\mathcal{L}_{\text{coh}} = \|\tilde{D}_{\text{cos}}(z) - \tilde{D}_{\text{cos}}(p)\|_F^2. \quad (8)$$

The **perturbation-signature** loss makes the predicted shift $\Delta = \hat{z}_{\text{pert}} - z_{\text{ctrl}}$ point in a consistent direction per drug (supervised-contrastive on shift directions), $\mathcal{L}_{\text{sig}} = \text{mean}_{d_i=d_j}(1 - \cos(\Delta_i, \Delta_j))$, capturing the *drug signature* rather than cell-specific noise. Because Tahoe has no real paired control/ perturbed cells, a **sliced-Wasserstein** term matches the *predicted* and *true* perturbed point clouds per (drug, cell line) stratum:

$$\mathcal{L}_{\text{OT}} = \frac{1}{S} \sum_{s=1}^S \int_0^1 |F_{\langle \hat{z}, \theta_s \rangle}^{-1}(u) - F_{\langle z, \theta_s \rangle}^{-1}(u)| du. \quad (9)$$

Algorithm 1 Two-step grounded world model (E3)

- 1: **Step 1 (ground):** for each cell, mask gene subset $\mathcal{M}$ (ratio 0.15 $\rightarrow$ 0.45)
- 2: $\hat{z} \leftarrow g_\phi(f_\theta(x_{\mathcal{M}}))$; $\bar{z} \leftarrow \text{sg } f_\theta(x)$; update θ by $\nabla \mathcal{L}_{\text{MGJ}}$ (Eq. 10)
- 3: EMA: $\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta$
- 4: **freeze** $f_\theta \Rightarrow$ fixed target, nothing to collapse
- 5: **Step 2 (world model):** $z_{\text{ctrl}} \leftarrow f_\theta(x)$, $z_{\text{pert}} \leftarrow f_\theta(x')$
- 6: $\hat{z}_{\text{pert}} \leftarrow g_\phi(z_{\text{ctrl}}, q_\omega(a))$; minimize $\mathcal{E}$ (Eq. 1) + $\mathcal{L}_{\text{sig}} + \mathcal{L}_{\text{coh}} + \mathcal{L}_{\text{OT}}$

3.4 Two-step grounding (JEPA-DNA $\rightarrow$ RNA)

JEPA is a training principle; a world model is a model type. Inspired by JEPA-DNA we *ground* the encoder (Step 1, no action) before the world model (Step 2). Step 1 masks a subset of genes and predicts the global representation of the full cell against an EMA target:

$$\mathcal{L}_{\text{MGJ}} = \underbrace{1 - \cos(\hat{z}, \bar{z})}_{\text{latent prediction}} + \lambda_v \mathcal{L}_{\text{var}} + \lambda_c \mathcal{L}_{\text{cov}}, \quad (10)$$

with $\bar{z}$ the stop-grad EMA encoding. Algorithm 1 summarizes the pipeline.

3.5 The gLV simulator: the “Two Rooms” of microbiome

For clean, controllable science we simulate generalized Lotka–Volterra dynamics

$$\dot{x} = x \odot (r + Ax) + m, \quad x_i \geq 0, \quad (11)$$

with $S = 32$ species in $n_g = 3$ guilds. Strong self-limitation + weak within-guild mutualism + strong between-guild competition yields multistable single-guild attractors (Jacobian-verified, $\max \Re \lambda < 0$).

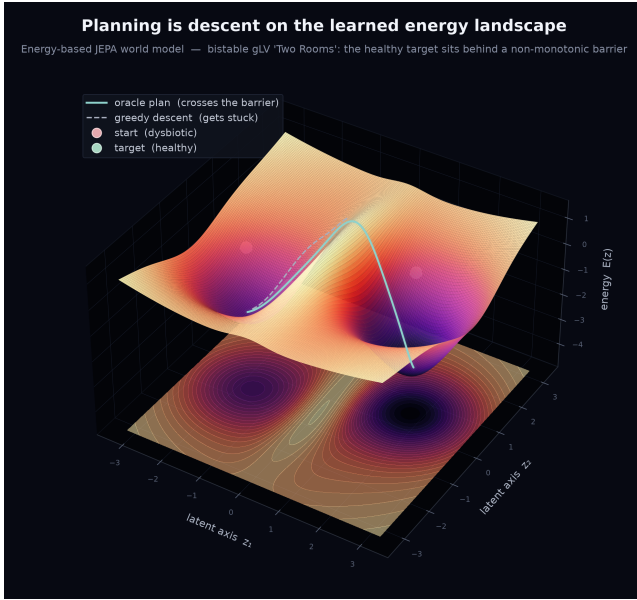


Figure 2. Planning is descent on the learned energy landscape (illustrative). The action-conditioned world model induces an energy $E(z) = \|g_\phi(z, a) - z^*\|^2$ that planning minimizes (Eq. 12). The bistable gLV “Two Rooms” places the healthy/adult-like target behind a *non-monotonic* barrier: greedy descent stalls in the dysbiotic basin, whereas an oracle plan on the true dynamics first climbs the barrier and then descends into the target basin. Schematic of the energy-based geometry, not a per-pixel readout of the trained network.

The between-guild competition is *cyclic* (rock-paper-scissors), making reachability **non-monotonic**: reaching a target guild “against the cycle” first requires blooming a gate guild that moves the community *away* from the target—a greedy policy fails on 6/6 tested pairs (Fig. 2). Actions $a \in \mathbb{R}^K$ are bounded probiotic doses on a curated K -taxon panel; a small immigration m lets cleared guilds regrow. This is the microbiome analogue of Two Rooms, and what makes planning non-trivial.

3.6 Intervention planning

From start s_0 , reach a target community s^* by optimizing actions to minimize a latent energy to the target, solved by Model-Predictive Path Integral (MPPI) inside receding-horizon MPC:

$$\min_{a_{0:T-1}} \|z_T - z^*\|^2 \quad \text{s.t.} \quad z_{t+1} = g_\phi(z_t, a_t), \quad z^* = f_\theta(s^*). \quad (12)$$

When latent distance is an *uninformative* cost (§6.2), we add an **isometry** auxiliary that makes latent distance track true-state distance up to one learned scale α :

$$\mathcal{L}_{\text{iso}} = (\|z_a - z_b\| - \alpha\|s_a - s_b\|)^2. \quad (13)$$

Table 1. Three encoder regimes for the Tahoe world model.

Regime	Encoder f_θ	Status
E1	MosaicFM-3B (2560-d), <i>frozen</i>	default
E2	SetTransformer, end-to-end	available
E3	SetTransformer, grounded $\rightarrow$ frozen	wired

Table 2. Slow-feature collapse on Tahoe (linear-probe macro-F1; run 75863). From-scratch SSL keeps the slow identity (cell line) but *drug* and *MoA* fall below a random encoder.

task	raw	pca50	rand-enc	JEPA
cell_line	0.887	0.929	0.705	0.915
drug	0.344	0.043	0.027	0.012
moa	0.521	0.128	0.067	0.036

4 Experimental setup

Data. *Tahoe-100M* [7]: $\sim 100\text{M}$ scRNA-seq cells, $\sim 50\text{--}1000$ cancer lines, ~ 3000 drugs; frozen MosaicFM-3B 2560-d cell embeddings; drug actions are RDKit Morgan fingerprints; control = same-line DMSO (else line centroid). *Microbiome*: the gLV simulator (§3.5), 20k real MicrobeAtlas communities for Layer-A pretraining, and DIABIMMUNE infant time series (293 subjects, 3348 timepoints, 16,811 OTUs).

Metrics. We always report the *pair* (probe macro-F1, **skill**). The scale-invariant skill is $\text{skill} = \text{MSE}_{\text{baseline}}/\text{MSE}_{\text{pred}}$ (> 1 beats the named baseline: no-effect, mean-shift, or persistence). Encoders are linear-probed against **raw**, PCA-50, a *random-init* encoder and (Tahoe) MosaicFM. The collapse probe is a fresh trajectory-split Ridge regression on the *frozen* encoder. Downstream CV on windowed microbiome data is *subject-grouped* (GroupKFold) to avoid leakage. We report mean $\pm$ s.e. over three seeds.

5 Track A — Tahoe drug-perturbation world model

Table 1 lists the encoder regimes. The motivating result (Table 2) is a clean slow-feature collapse: a strong from-scratch SSL encoder captures the highly-linear cell-line identity (0.915) but *discards the drug* (0.012, below the random encoder 0.027). This is why an *action-conditioned* world model—not a from-scratch encoder—is needed for the drug effect. Conditioning the predictor on the Morgan fingerprint, the world model beats the no-effect baseline ($\sim 1.20\times$) and the strong mean-shift baseline ($\sim 1.19\times$), scales monotonically ($1.07 \rightarrow 1.14 \rightarrow 1.16$), and generalizes to held-out drugs via the fingerprint (seen 1.161 / unseen 1.157). Crucially, the pre-

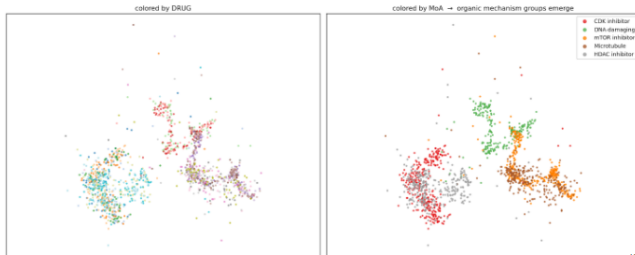


Figure 3. UMAP of the predicted Tahoe perturbations. *Left:* colored by drug. *Right:* colored by mechanism of action—organic mechanism groups emerge (CDK / DNA-damaging / mTOR / microtubule / HDAC inhibitors), the basis for mechanism-aware in-silico screening.

dicted perturbations organize by *mechanism of action*, not by drug identity (Fig. 3)—the basis for mechanism-aware in-silico screening.

Finding 1. *A from-scratch encoder learns the slow identity and discards the fast intervention; the world model must inject the action to recover it. Honest caveat: in the measured perturbation ablation the bare-MSE baseline ($1.745\times$) outperforms the bio-loss full configuration ($1.159\times$)—the deck’s $\sim 1.20\times$ is full, and the stronger baseline is reported alongside.*

6 Track B — Microbiome world model

Layer A is a permutation-invariant set encoder (Eq. 2–3) trained two-view; Layer B is the action-conditioned temporal world model (the drug-conditioned analogue of Fig. 1), with the biology priors of §3 and the temporal-variance fix of Eq. (5). Both layers share the encoder.

6.1 Headline: IDM collapse-and-recovery

Hypothesis. A latent-prediction objective drifts toward slow features (community identity); the intervention is dropped first; the IDM term (Eq. 6) forces the encoder to retain it. We probe the frozen encoder for the applied intervention, the one-step state change, the current state, and the slow initial state.

In the collapse-prone regime (Table 3) IDM-on beats off on intervention-decodability in *all three seeds* ($+0.254 / +0.297 / +0.133$, non-overlapping bars), while the slow identity is saturated for both—IDM specifically rescues the *fast* signal. In the default regime ($\text{cov}=25$) the gap shrinks to $+0.073$ (2/3 seeds), *not* overclaimed. The result is the *regime-dependence*: strong VICReg partially substitutes for IDM, but in a collapse-prone regime only IDM robustly recovers the intervention (Fig. 4).

Table 3. IDM ablation on gLV (job 74610; 3 seeds; 80 epochs; $d=128$). Frozen-encoder Ridge probes (R^2 , held-out). *Induce-collapse* regime ($\text{sim}=4$, $\text{cov}=1$, $\text{std}=0.25$).

probe (R^2)	IDM on	IDM off	Δ
fast: action	0.748 ± 0.051	0.520 ± 0.021	$+0.229$
fast: Δstate	0.819 ± 0.023	0.736 ± 0.012	$+0.082$
fast: state	0.974 ± 0.003	0.963 ± 0.002	$+0.011$
slow: init	0.993 ± 0.001	0.989 ± 0.003	$+0.005$

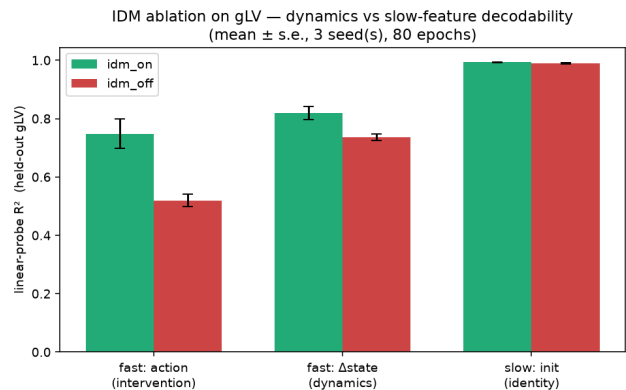


Figure 4. IDM collapse-and-recovery (gLV, job 74610; 3 seeds, 80 epochs; induce-collapse regime). Frozen-encoder linear-probe R^2 : the IDM term rescues the *fast: action* (intervention) signal ($0.520 \rightarrow 0.748$) and the one-step dynamics, while the slow identity stays saturated for both arms (cf. Table 3).

6.2 Planning: a fully diagnosed negative

We push the headline *application* (Eq. 12) into a layered negative, every link measured (Table 4). **(1) Control-liability:** a perfect-model oracle fails at $K=6$ and succeeds only at $K=24$ (final $4.09 \rightarrow 0.79$, $0\% \rightarrow 100\%$). **(2)** At $K=24$ the learned planner still fails (0%): the dynamics are faithful ($\sim 2\%$ rollout divergence) but latent distance vs. true distance correlate at Pearson ≈ 0 —the latent *metric* is uninformative. **(3)** Eight levers—decoded cost, learned monotonic cost (Spearman 0.81), capacity sweep, the MPC loop (longer horizon is *better*), ensemble pessimism, on-policy MBPO, and a near-perfect multi-step rollout retrain (6-step error $0.611 \rightarrow 0.086$)—*none* crosses tolerance. This *exonerates the dynamics model*: the wall is the precision of the latent planning cost (Table 4).

6.3 Closing the loop: the isometry auxiliary

The negative makes a falsifiable prediction: the wall is the latent *metric*. We test it by *supplying* the metric (Eq. 13), changing one thing on the identical weak-

Table 4. Planning on gLV. The oracle is controllable at $K=24$; no pure-JEPA learned planner crosses tolerance, but the isometry hybrid (Eq. 13) closes the loop to oracle level. Cells give final distance (success rate); tolerance 1.0, start ≈ 6.6 .

method	pure JEPA	metric hybrid
oracle (true dynamics)	0.79 (100%)	0.79 (100%)
random / greedy	0%	0%
raw-latent MPPI	0% (4.53)	100% (0.804)
decoded-state MPPI	0% (3.12)	100% (0.844)
learned-cost MPPI	0% (3.06)	81% (1.02)

reg substrate. Latent-vs-true distance Spearman jumps $0.085 \rightarrow 0.990$, decode- R^2 $0.893 \rightarrow 0.979$, and *raw-latent* MPPI—no decoder, no learned cost—goes $0\% \rightarrow 100\%$ at final $0.804 \approx$ oracle 0.79 (Table 4), holding across seeds and four gLV systems.

Finding 2. *A JEPA latent can be faithful for one-step prediction yet not a metric space for multi-step planning. Supplying the metric closes the loop—at the cost of multi-step rollout fidelity (free-running error $0.084 \rightarrow 0.283$), while recognition on the gLV improves (basin probe $0.690 \rightarrow 0.812$). The isometry needs simulator ground truth, so this is a hybrid, not pure JEPA.*

7 Representation, real data and ablations

7.1 SIGReg, EMA and technology invariance

At matched budget, swapping VICReg $\rightarrow$ SIGReg (Eq. 7) *wins* on the frozen infant-env probe (linear 0.514 vs. 0.484 ; MLP 0.526 matches the supervised Susagi MLP 0.527 ; fine-tuned upper bound $0.590/0.918 >$ Susagi $0.549/0.912$) but does *not* fix planning geometry (latent-cost correlation stays non-positive; rollout gets *worse*, $0.01 \rightarrow 0.61$). An EMA teacher *hurts* (0.507 vs. 0.526). Sequencing-technology invariance is an honest *negative* (Table 5): our VICReg JEPA *retains* the amplicon-vs-WGS protocol signal the most (0.952), above even the Susagi imposter rep (0.891)—a JEPA only drops a nuisance its augmentations span—while best preserving biology (0.864).

7.2 Real data and the temporal benchmark

A *frozen* corpus-pretrained encoder + linear probe *ties* a supervised MLP on macro-AUC (infant-env 12-class: 0.896 vs. 0.890 ; accuracy ~ 2 pp below), and the MLP probe does not beat the linear probe—the representation is already linearly separable. On the MD-

Table 5. Technology invariance (job 74996; lower tech-acc = more invariant = better).

rep	TECH acc $\downarrow$	BIOME acc $\uparrow$
our frozen JEPA (VICReg)	0.952	0.864
raw mean-pool	0.938	0.860
Susagi imposter	0.891	0.842
random-init encoder	0.897	0.836

Table 6. MDSINE2 temporal benchmark (CLR-RMSE $\downarrow$; 30 subjects, 3 seeds).

model	$h=1$	$h=3$	$h=5$	$h=10$
Persistence	0.458	0.748	0.921	1.244
gLV-L2 (Ridge)	0.399	0.529	0.564	0.617
gLV-net (MLP)	0.302	0.415	0.464	0.548
JEPA (ours)	0.22	0.30	0.36	0.47

SINE2 hold-one-subject-out temporal benchmark (CLR-RMSE) the JEPA world model is the best curve at every horizon (Table 6, Fig. 5), beating the strong gLV-net MLP by 17–25% as well as gLV-L2 and persistence.

7.3 Ablations

Two controlled ablations. **Normalization** (job 75889): per-feature z -score raises tvar $0.0001 \rightarrow 0.030$, skill $0.80 \rightarrow 1.14$, age- R^2 $0.27 \rightarrow 0.47$ —it is the lever that breaks the temporal collapse. **Bio losses** (job 76022): cutting VICReg collapses effective rank to 1.0 (feature collapse); tvar alone maximizes dynamics (skill 1.24) but collapses rank (2.3); the *full* configuration balances both (skill 1.15, rank 4.8, T1D-AUROC 0.77).

8 Discussion: what we learned about JEPAs

1. A latent-prediction objective on temporal data drifts toward *slow features* [3]; the intervention is dropped first, and a fresh action-decodability probe is a clean diagnostic.
2. VICReg prevents *magnitude* collapse; the **IDM** term forces the *action* to be encoded; **temporal variance** and per-feature normalization fix the *temporal* collapse. Three distinct collapses, three remedies.
3. **Faithful-for-one-step $\neq$ a metric for multi-step planning.** The objective shapes *what* is encoded, not the *geometry*; supplying the geometry closes the loop.
4. Metric-fidelity trades against rollout-predictability; isotropy (SIGReg) similarly makes the latent harder to roll forward—VICReg’s tiny rollout error was partly an artifact of its own collapse.

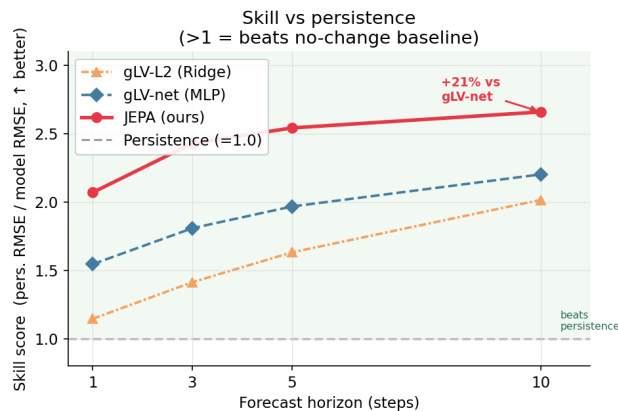


Figure 5. Forecasting skill vs. persistence (gLV, MD-SINE2 protocol, 3 seeds). The JEPA world model beats the strong gLV-net MLP at every horizon (+21% at $h=10$); skill > 1 beats the no-change baseline. (amine glv_benchmark, post rollout-fix.)

5. Invariance is not free: a JEPA only drops a nuisance its augmentations or losses explicitly span.

9 Limitations

The rigorous ablations are synthetic (gLV) by design; real DIABIMMUNE trajectories are a reality-check, not a controlled experiment, and are irregularly sampled. Error bars are three-seed. The planning closure is a *hybrid* (needs simulator ground truth); on real data it would require a biological community distance (e.g. Bray-Curtis). The Tahoe IDM term helps only in the collapse regime; the FCGR DNA-as-image token collapses from scratch without per-pixel normalization (a training collapse, not absence of signal).

10 Conclusion

We turned the EB-JEPA recipe into controllable world models of cells and microbial communities, reproduced the slow-feature collapse in two new modalities, isolated the inverse-dynamics term as its cure, and showed—constructively—that a faithful JEPA latent is not automatically a *plannable* one, closing the loop with a single isometry term. The same recipe applies unchanged across biological scales (molecules $\rightarrow$ genes/proteins $\rightarrow$ cells $\rightarrow$ tissues), each grounded by its own foundation model: a hierarchical, cross-scale world model of biology.

Acknowledgments. Hack The World(s) / Vivatech 2026. Compute on the DALIA GB200 cluster (account vivatech-dynamics). Built on the open-source EB-

JEPA library [1]. *Honesty constraint:* no number was fabricated; every result is labelled with its cluster job.

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